

Technical Document #877: Deep Learning - Comprehensive Overview

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Recurrent neural networks (RNNs) are designed to process sequential data. They maintain hidden states that capture information about previous elements in the sequence.

Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies. They use gating mechanisms to control the flow of information.

Generative adversarial networks (GANs) consist of two neural networks competing against each other. The generator creates synthetic data while the discriminator evaluates authenticity.

Batch normalization normalizes the inputs of each layer, reducing internal covariate shift. It allows higher learning rates and acts as a regularizer.

Convolutional neural networks (CNNs) are designed to process grid-like data such as images. They use convolutional layers to automatically learn spatial hierarchies of features.

Transfer learning is a technique where a model trained on one task is re-purposed on a second related task. It has revolutionized computer vision and NLP.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Key Takeaways:

- Generative adversarial networks (GANs) consist of two neural networks competing against each other.
- Batch normalization normalizes the inputs of each layer, reducing internal covariate shift.
- Dropout is a regularization technique where randomly selected neurons are ignored during training.